FACULDADE DE INFORMÁTICA E ADMINISTRAÇÃO PAULISTA (FIAP)

CHALLENGE Natura SPRINT 1

SÃO PAULO – SP

2023

BACHARELADO EM ENGENHARIA DA COMPUTAÇÃO – 1º SEMESTRE

ANÁLISE DE PROBLEMAS MULTIVARIÁVEIS

Bruno Tanaui Zanocco - RM: 85770

Caio Silva Alves - RM: 85350

Guilherme Macario da Silva - RM: 4057

Kaique Carvalho da Silva - RM: 93040

Murilo José Cressoni - RM: 95267

SÃO PAULO -SP 2023

$C'(t) = 30 - 0{,}06t$

$C(150) = 34940$ $\qquad 0 \leq t \leq 1725$

Questão 1

$$C(t) = \int C'(t)\, dt$$

$$C(t) = \int (30 - 0{,}06t)\, dt$$

$$C(t) = \int 30\, dt - 0{,}06 \int t\, dt$$

$$C(t) = 30t - 0{,}03t^2 + c^{\bullet}$$

$C(150) = 35450$

$30t - 0{,}03t^2 + c^{\bullet} \Rightarrow t = 150$

$C = 35450$

$30(150) - 0{,}03(150)^2 + c^{\bullet} = 35450$

$c^{\bullet} = 35450 - 3(150) + 0{,}03(150)^2$

$c^{\bullet} = 35450 - 1350 + 675$

$c^{\bullet} = 34775$

1) Dessa forma

$$C(t) = 90t - 0{,}03t^2 + 34775 \;//$$

Item 2

$$C(t) = 90t - 0{,}03t^2 + 34775$$

$$C(0) = 90(0) - 0{,}03(0)^2 + 34775$$

$$C(0) = 34775 \;//$$

Item 3

$C(1000)$

$$C(t) = 90t - 0{,}03t^2 + 34775$$

$$C(1000) = 90(1000) - 0{,}03(1000)^2 + 34775$$

$$C(1000) = 90000 - 30000 + 34775$$

$$C(1000) = 94775 \;//$$

Questão 4

a) $0 < t < 1725$

3 ganhos para fazer o gráfico

$C(t) = 90t - 0{,}03t^2 + 34775$.

$t = \frac{-b \pm \sqrt{\Delta}}{2a}$ $\quad \Delta = 90^2 - 4(-0{,}03)(34775)$

$\Delta = 12273$

$t = \frac{-90 \pm \sqrt{12{,}273}}{2(-0{,}03)}$

$t' = \frac{-90 + 110{,}7836}{-0{,}06} = -346{,}3929$

$t'' = \frac{-90 - 110{,}7836}{-0{,}06} = 3346{,}3929$

Ótimo 7mm 2 ganhos

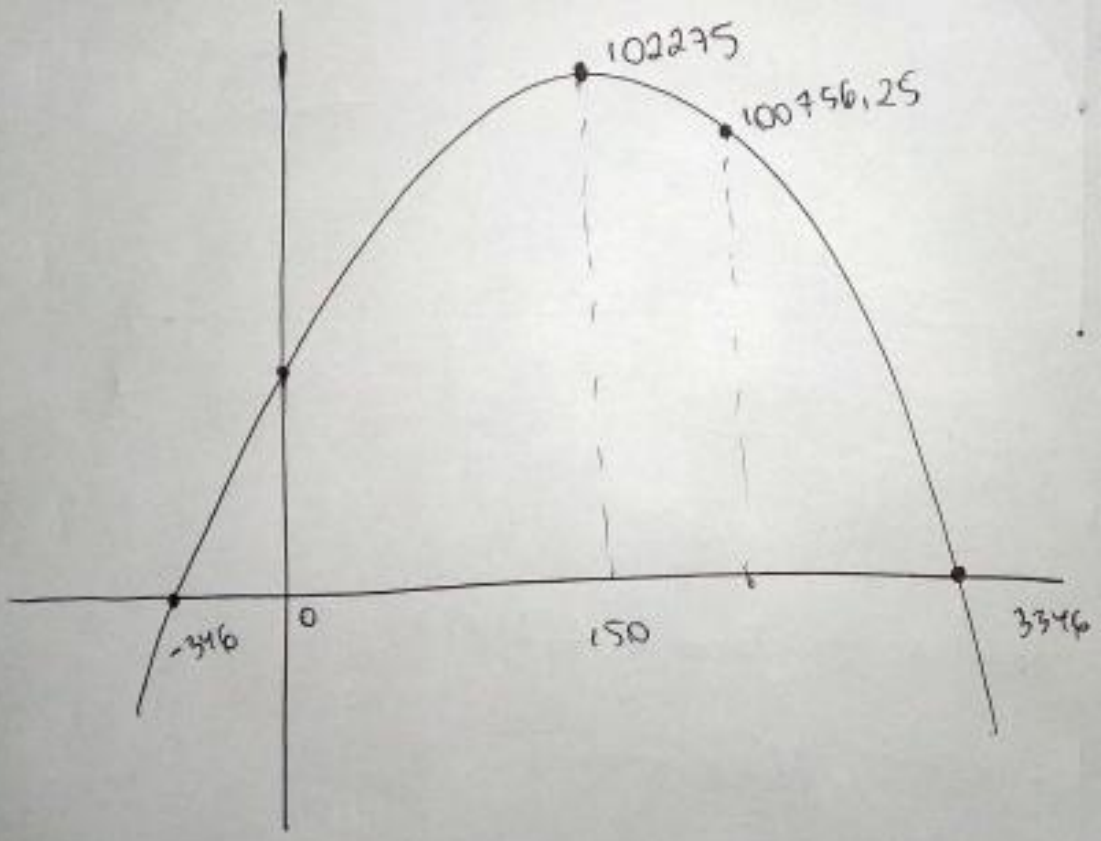
Tuvo 4 ganchos

(0, 34775)

(-346,3929 , 0)

(3346, 3929, 0)

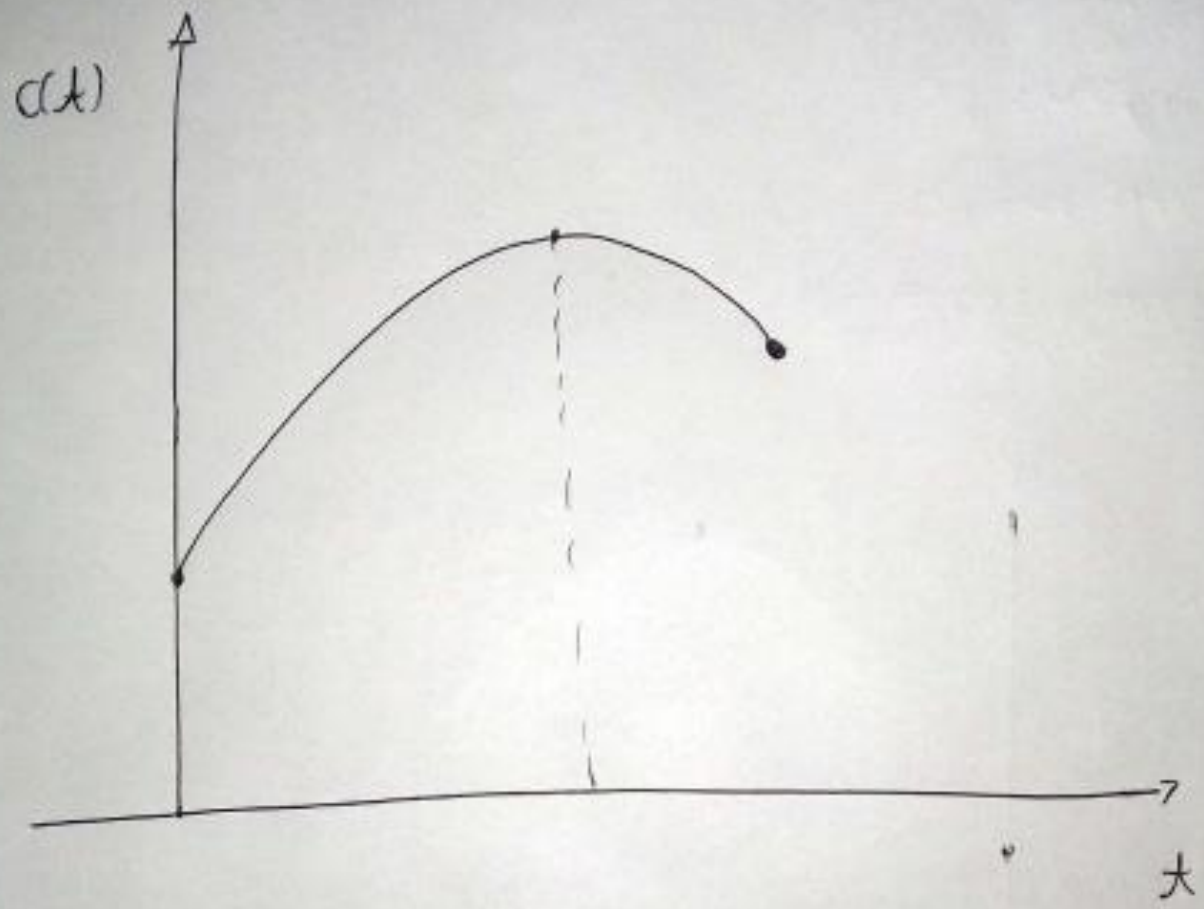
(150 , 102275)



$C(1725) = 90(1725) - 0,03(1725)^2 + 34775$

$C(1725) = 100756.25$

Mas o intervalo é $0 \leq t \leq 1725$

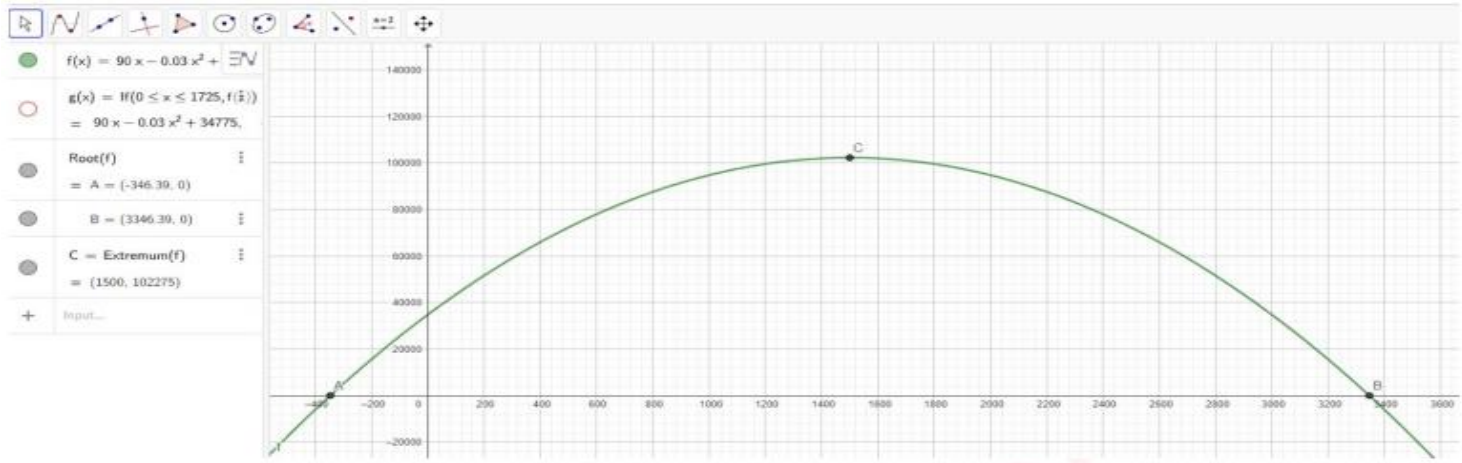


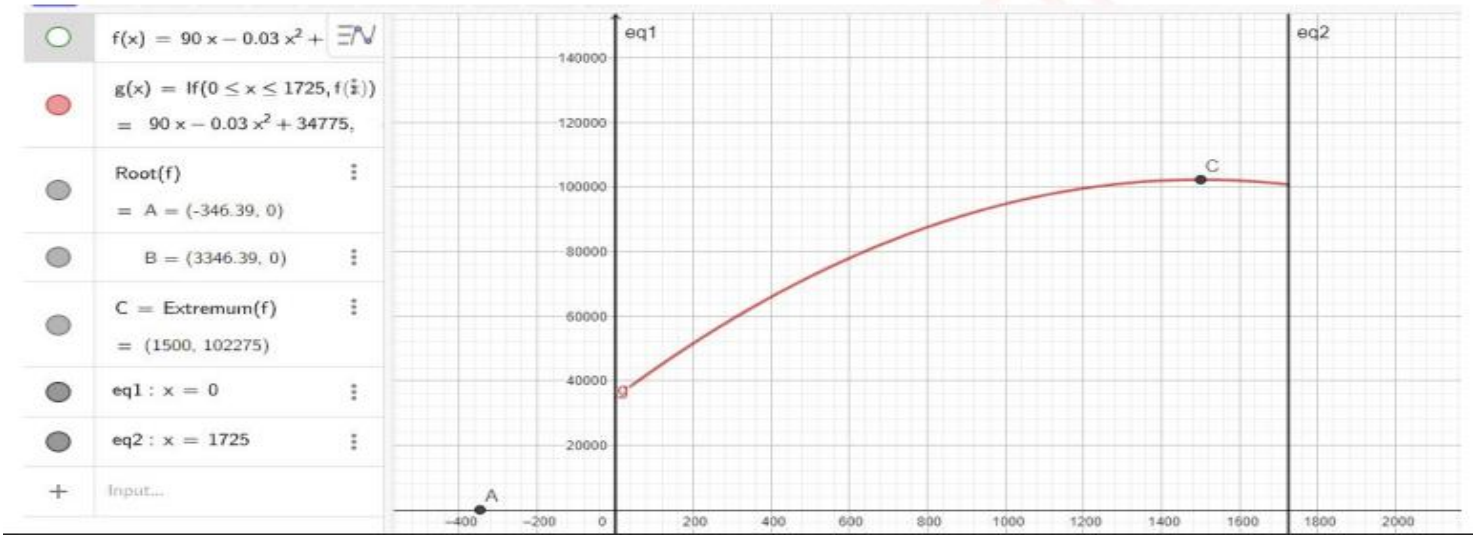
Assim

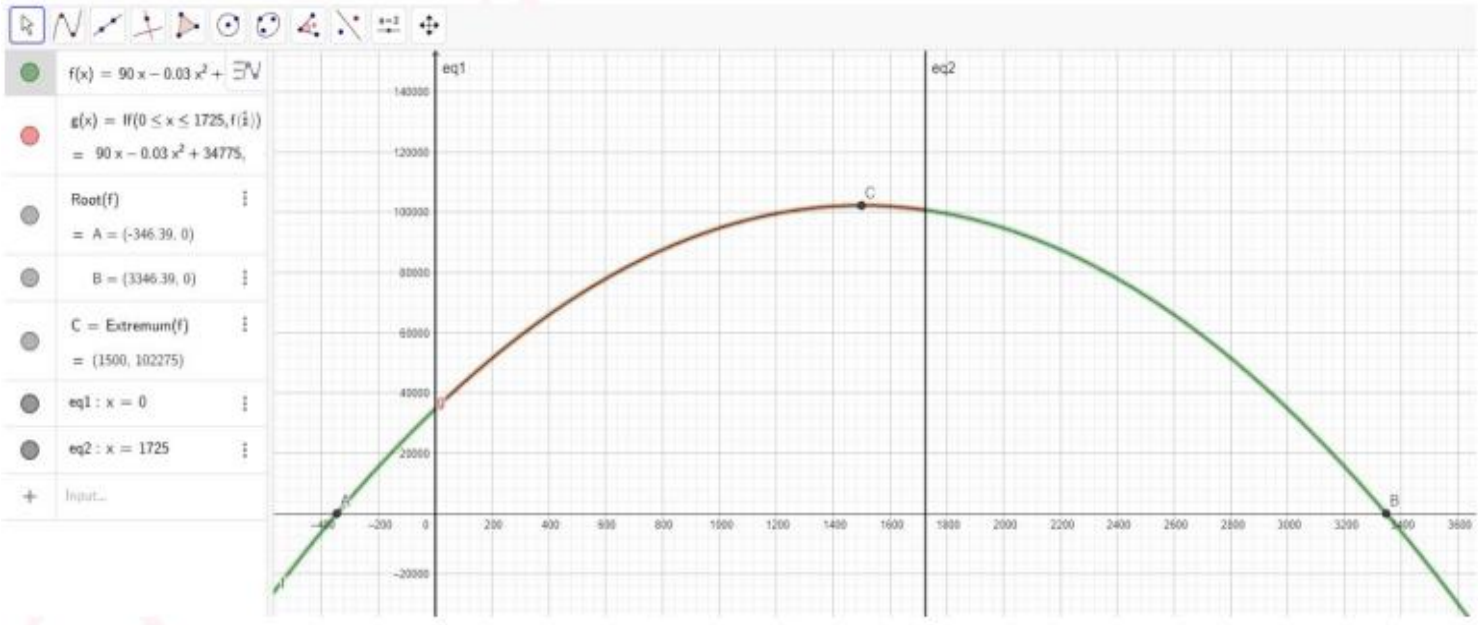
Crescimento $0 \leq t < 150$

$t = 150 \rightarrow$ Máximo

Decrescimento $150 < t \leq 1725$ //


f(x) = 90 x − 0.03 x² +
g(x) = If(0 ≤ x ≤ 1725, f(x))
= 90 x − 0.03 x² + 34775,
Root(f)
= A = (-346.39, 0)
B = (3346.39, 0)
C = Extremum(f)
= (1500, 102275)
Input...
A
B
C


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= (1500, 102275)
eq1 : x = 0
eq2 : x = 1725
Input...
eq1
eq2
A
C


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